Impact of Data Analytics on Consumer Buying Behaviour: The Role of Predictive Analytics, Recommendation Systems, Targeted Advertising, and Data Privacy

Bharath R

Assistant Professor
Post Graduate Department of Business Administration

Research Scholar, Seshadripuram First Grade College, Bengaluru

Email: bharathgowda.sdc@gmail.com

Mohan Kumar G

Post Graduate Department of Business Administration

Student, Seshadripuram College, Bengaluru

Email: mohankumargajendran2825@gmail.com

Vignesh V
Post Graduate Department of Business Administration

Student, Seshadripuram College, Bengaluru
Email: Vignesh.v1313@gmail.com

ABSTRACT

The rapid growing of digital technologies has transformed the way businesses understand and engage with their consumers. In today's competitive marketplace, data analytics has become an essential tool for analyzing customer preferences, their purchasing patterns and the ongoing market trends, which helps in enabling organizations to make informed and strategic marketing decisions. The primary objective of this study is to examine the impact of the data analytics on consumer buying behavior, with particularly, highlighting on their purchasing decisions, customer satisfaction and the brand loyalty. A quantitative research approach was carried out and primary data was collected from 210 respondents using a structured questionnaire. The collected responses were later analyzed to examine the relationship between data analytics practices and consumer purchasing behavior. This study also explores the role of a predictive analytics, their recommendation systems and the targeted advertising while identifying the importance of data privacy and the ethical use of consumer information. The findings are expected to provide valuable insights that can help the businesses design more consumer centric marketing strategies, improve their decision making and strengthen consumer trust in an increasingly data driven business environment.

Keywords: *Data Analytics, Consumer Buying Behaviour, Predictive Analytics, Recommendation Systems, Targeted Advertising, Data Privacy*

INTRODUCTION

Over the last few years, the way the business organisations interact with their consumers has changed notably. Because of the increase in the usage of shopping websites, social networks, mobile apps and digital payment systems, consumers generate substantial amount of information via their regular web interactions. Business organisations leverage this information in order to get insight into the consumers' preferences, understand the purchasing patterns and make marketing decisions based on the obtained data. Thus, data analytics is becoming a key component of any marketing strategy planning and customer relationships management process.

The increasing utilisation of the digital technology has changed the consumers' expectations on the businesses. Today's consumers no longer site their decisions simply on factors such as price, product quality or the brand reputation. They want personalised experience, appropriate recommendations, communication and services that correspond their preferences. As the consumers' expectations are evolving with time, the businesses are also seeking out strategies to utilise with the use of technologies such as predictive analytics, recommendation systems, machine learning, and targeted advertising to give out more personalised and relevant experiences.

However, while these technological advancement have created new opportunities and pathways for the businesses entitles, they have also highlighted to handle the consumer data responsibly with its growing needs. And as Consumers are becoming more and more aware of the security and the privacy concern, business firms are expected to follow marketing strategies that are more effective, consumer focused and ethically responsible.

Research Methodology

The study will engage on a quantitative research design to examine the impact of the data analytics on consumer buying behaviour. The primary data will be collected through a structured questionnaire conducted with along 210 participants. The questionnaire will be distributed and completed manually to obtain the required information. A convenience sampling techniques will be used to collect data from participants who are easily available and suitable for the study. The survey aims to examine how data analytics influence consumer on their purchasing decisions and to help better understand its role in shaping their buying behaviour. In addition, this survey will be carried out to establish the extent to which consumers are influenced by such aspects as predictive analytics, recommendation models, targeted advertisements, and data privacy. As a result, the data collected will be analysed using descriptive statistics to describe the general features of the data sample and conclude the research question and hypothesis. On the other hand, statistical analysis will be done to assess the relationship between the independent variable, that is, data analytics, and consumer buying behaviour. The study examines the relationship between data analytics, the independent variable, and the consumer buying behaviour. It aims to provide a better understanding of how data analytics influences consumers purchasing decisions and offers a better understanding of how data analytics influence consumer behaviour.

Data Analysis and Interpretation

Variable	Category	Frequency (N)	Percentage (%)
Gender	Male	110	51.89
	Female	102	48.11
Age	Below 20 years	56	26.42
	21–30 years	140	66.04
	31–40 years	16	7.55
Educational Qualification	SSLC/PUC	20	9.43
	Undergraduate	102	48.11
	Postgraduate	82	38.68
	Others	8	3.77
Occupation	Student	138	65.09
	Private Employee	42	19.81
	Business/Self-employed	20	9.43
	Government Employee	4	1.89
	Studying Fashion Designing	2	0.94
	Housewife	2	0.94
	Homemaker	2	0.94
	Teaching	2	0.94
Monthly Income	Below ₹20,000	154	72.64
	₹20,001–₹40,000	34	16.04
	₹40,001–₹60,000	18	8.49
	Above ₹60,000	6	2.83
Frequency of Online Shopping	Frequently	54	25.47
	Occasionally	88	41.51
	Rarely	66	31.13
	Never	4	1.89
Preferred Online Shopping Platform	Amazon	96	45.28
	Flipkart	60	28.30
	Meesho	24	11.32
	Myntra	22	9.43
	Instagram	4	1.89
	None of the Above	2	0.94
	Flipkart & Ajio	2	0.94
	D Mart	2	0.94

The demographic profile indicates that most respondents were male (51.89%) and belonged to the 21–30 years age group (66.04%). Nearly half were undergraduates (48.11%), while students constituted the largest occupational group (65.09%). Most respondents had a monthly income of less than ₹20,000 per month (72.64%) and shopped online occasionally (41.51%). Amazon emerged as the most preferred online shopping platform (45.28%). Overall, the sample primarily represents young, educated, and active online shoppers.

Scale Reliability Statistics			
Constructs	Mean	SD	Cronbach's α
Predictive Analytics	3.52	0.565	0.727
Recommendation System	2.89	0.551	0.789
Targeted Advertisement	3.41	0.635	0.727
Data Privacy	3.58	0.565	0.749
Consumer Buying Behaviour	3.44	0.511	0.727

The reliability analysis demonstrates satisfactory internal consistency for all constructs. Cronbach's alpha values ranged from **0.727 to 0.789**, exceeding the recommended threshold of **0.70**. Recommendation System recorded the highest reliability (α = 0.789), while Predictive Analytics, Targeted Advertisement, Data Privacy, and Consumer Buying Behaviour also showed acceptable reliability. These findings confirm that the measurement instrument is reliable and suitable for subsequent statistical analyses.

Corelation

Correlation Matrix						
		Predictive	RS	TAP	DP	CB
Predictive Analytics	Pearson's r	—				
	df	—				
	p-value	—				
Recommendation System	Pearson's r	0.585	—			
	Df	210	—			
	p-value	<.001	—			
Targeted Advertisement	Pearson's r	0.473	0.542	—		
	Df	210	210	—		
	p-value	<.001	<.001	—		
Data Privacy	Pearson's r	0.456	0.582	0.415	—	
	Df	210	210	210	—	
	p-value	<.001	<.001	<.001	—	
Consumer Behaviour	Pearson's r	0.551	0.664	0.594	0.619	—
	Df	210	210	210	210	—
	p-value	<.001	<.001	<.001	<.001	—

Interpretation

The correlation analysis indicates that all independent variables have positive and statistically significant relationships with Consumer Buying Behaviour ($p < 0.001$). Recommendation System exhibited the strongest correlation ($r = 0.664$), followed by Data Privacy ($r = 0.619$), Targeted Advertisement ($r = 0.594$), and Predictive Analytics ($r = 0.551$). These findings suggest that improvements in data analytics practices are associated with higher levels of consumer buying behaviour.

Linear Regression

Model Fit Measures		
Model	R	R^2
1	0.769	0.591
Note. Models estimated using sample size of N=212		

The regression model produced an **R value of 0.769** and an **R^2 value of 0.591**, indicating a strong relationship between the independent variables and Consumer Buying Behaviour. The model explains **59.1%** of the variation in Consumer Buying Behaviour, while the remaining **40.9%** is attributed to other factors not included in the study. This demonstrates that the model has good explanatory power.

Model Coefficients – Consumer Behaviour				
Predictor	Estimate	SE	t	P
Intercept	0.487	0.1755	2.77	.006
Predictive Analytics	0.122	0.0515	2.36	.019
Recommendation System	0.249	0.0577	4.32	<.001
Targeted Advertisement	0.210	0.0440	4.78	<.001
Data Privacy	0.263	0.0507	5.19	<.001

The regression results reveal that all four independent variables significantly and positively influence Consumer Buying Behaviour. Data Privacy ($\beta = 0.263$) exerted the strongest effect, followed by Recommendation System ($\beta = 0.249$), Targeted Advertisement ($\beta = 0.210$), and Predictive Analytics ($\beta = 0.122$). Since all p-values are below 0.05, each predictor significantly contributes to explaining consumer buying behaviour.

Hypothesis	**Path**	**β (Estimate)**	**t-value**	**p-value**	**Decision**
H1: Predictive Analytics has a significant positive influence on Consumer Buying Behaviour.	PA → CBB	0.122	2.36	0.019	**Accepted**
H2: Recommendation System has a significant positive influence on Consumer Buying Behaviour.	RS → CBB	0.249	4.32	<0.001	**Accepted**
H3: Targeted Advertisement has a significant positive influence on Consumer Buying Behaviour.	TA → CBB	0.210	4.78	<0.001	**Accepted**
H4: Data Privacy has a significant positive influence on Consumer Buying Behaviour.	DP → CBB	0.263	5.19	<0.001	**Accepted**

Discussion:

The results of this study demonstrate that data analytics significantly contribute to shaping consumer behaviour. The four variables, including predictive analytics, recommendation system, targeted advertising and data privacy, positively and substantially influenced the consumers buying behaviour. The recommendation system appeared to have the highest correlation, while the data privacy presented the highest beta coefficient. Overall, the developed model could

explain nearly 60% of variance in consumers buying behaviour, which demonstrates a strong connection between the independent variables and the dependent one. The results of the study suggest that providing consumers with tailored recommendations and relevant advertising could boost purchase intentions, while targeted data analytics and privacy policies increase the level of trust and confidence among customers. Therefore, organisations must consider data analytics as a vital tool for influencing consumer behaviour, implementing personalization as well as remaining transparent and ethical with consumer's data.

Conclusion:

This study hypothesized that data analytics would have an impact on the consumer buying behavior. This impact is manifested through the application of predictive analytics, recommendation systems, target advertising, and the relevance of data privacy. The hypothesis has been supported by the results obtained from the study. First, the study found out that data privacy had a great effect on consumer buying behavior. Therefore, this study would advise consumers to demand more data security and transparency in order to obtain value from their personal data. Secondly, recommendation systems had an effect on consumer buying behavior by introducing convenience in shopping. Thirdly, there was a positive relationship between consumer buying behavior and predictive analytics. The results also found out that the research model accounted for a reasonable percentage of the variance in consumer buying behavior. These results indicate some of the ways that consumer buying behavior can be influenced, including the need for personalization while ensuring confidentiality. The study suggests that organisations should implement in data analytics in a way which helps in promoting transparency, protects the consumer's data and strengthens the consumers trust. By using data analytics responsibly, Businesses can gain a better understanding of how their consumers choose and also respond to their changing expectations and built a lasting relationship with them through trust.

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